

BRICK AND STONE

First appearing around 3000 BCE, robust fired bricks facilitated the development of new forms of architecture such as arches, vaults, and domes. However, stone has greater compressive strength than brick—and, once tools had been invented to cut it, buildings began to soar.

Early stone structures

Evidence survives for early stone structures erected during the Neolithic Age (10000–3000 BCE), such as the Megalithic Temples of Malta (see p.94), dated about 3700 BCE, and the Cairn of Barnenez in France, dating from 4850 BCE. These are relatively crude structures of piled, undressed stones. The advanced civilizations in the Nile valley were the first to develop the ability to cut and shape stone, and thus to build structures of immense size and durability. Their first great monument was the Stepped Pyramid of Djoser (2667–2648 BCE)—the earliest known large-scale cut-stone construction. This was followed roughly 100 years later by the Great Pyramid of Giza (see pp.208–209), whose visual impact remains stunning. The Egyptians' building techniques advanced quickly, and by 2000 BCE, they were creating vast temple complexes with pylons, courtyards, colonnades, and huge roofed halls. The culmination was Karnak (in modern-day Luxor; see pp.210–211), the largest religious complex ever known, built on such a scale that its columns rose up to 80 ft (24 m) high.

Greek and Roman architecture

Across the Mediterranean, the Greeks adopted the same stone-frame temple construction form as the Egyptians—column-and-beam (or post-and-lintel). They brought to building in stone a new refinement and a set

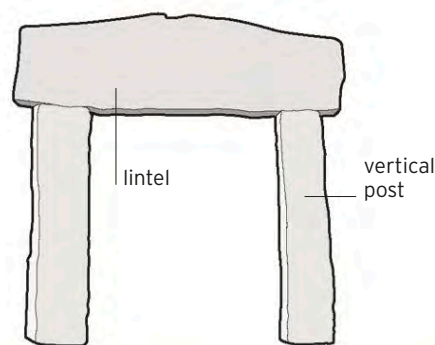


◁ INTRICATE DECORATION

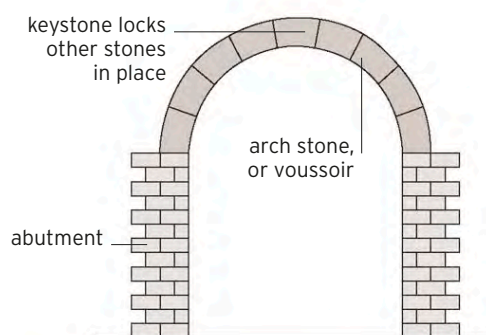
Around the same time that Gothic cathedrals were spreading across Europe, the Khmer empire (802–1431) in Southeast Asia was constructing temple complexes of great elaboration out of brick and sandstone.

of aesthetics and architectural orders that are still adhered to, notably Doric, Ionic, and Corinthian (see p.97). They also developed the bas-relief decoration of the Egyptians into full-blown sculpted friezes and introduced optical corrections to show their buildings to best advantage.

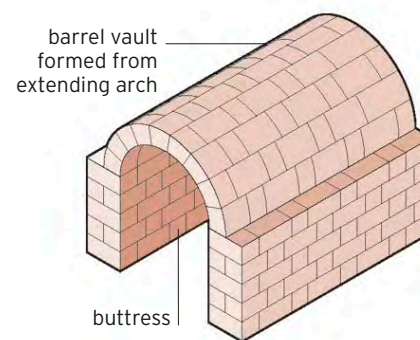
It was the Roman builders, however, who exploited masonry to its fullest advantage. Archaeological digs have revealed the first true arches in brick and the earliest forms of vaulting in the sewers and tombs of Mesopotamia. It is clear, then, that the Romans did not invent



POST-AND-LINTEL



ARCH



VAULT



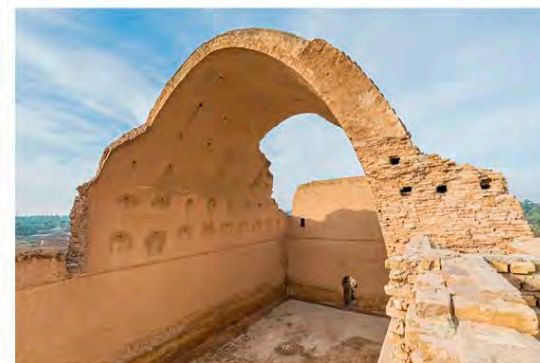
△ POST-AND-LINTEL CONSTRUCTION

Some of the earliest forms of stone architecture were post-and-lintel structures, with horizontal blocks supported on vertical posts. The Egyptians and Greeks refined this into columns, entablatures, and pediments, as seen in the Parthenon temple in Athens (see pp.98–99).



△ ARCHES

The invention of the arch enabled builders to span far larger openings than had been possible previously. The Romans were masters of arch building, as in this aqueduct, the Pont du Gard at Nîmes in France (see p.101).



△ VAULTING

From the arch came the basic barrel or tunnel vault. This could be used to create vast internal spaces, as at Ctesiphon in Iraq. Later, vaulting developed into ever more elaborate forms, such as the intersecting groin vault.